Progressive Learning for Unsupervised Change Detection on Aerial Images

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***Abstract*— This article focuses on unsupervised methods for optical aerial image change detection. Existing unsupervised change detection techniques are mainly categorized as patch-based methods and transfer-learning-based methods. However, the first type ignores the spatial information in the images, and the second type may introduce new errors due to knowledge extracted from additional datasets. To effectively tackle these problems, we propose an unsupervised progressive learning framework (UPLF). We first use original estimated change maps as the labeled samples and choose the reliable regions from samples to train the network. We then propose a progressive learning method to expand the reliable labeled region. Briefly, we apply a label selection filter to filter out incorrect change information from the regions to help rectify incorrect labeling in the regions. This leads to a more reliable labeled region and thus, in turn, more accurate detection results. Compared with the patch-based and transfer-learning-based unsupervised techniques, our method takes the entire map as the training sample to avoid the problem associated with using small patches; moreover, our iterative and progressive methods further enhance the change detection performance without involving external knowledge. Indeed, based on our experimental results on the real datasets, the proposed method demonstrates highly competitive performance compared with the state-of-the-art.**

***Index Terms*— Change detection, convolutional neural network (CNN), optical aerial images, unsupervised learning.**

I. Introduction

WITH the rapid development of aviation technologies (e.g., unmanned aerial vehicles), many optical aerial images having high ground resolution and rich content can be acquired [1], [2]. This makes possible the advanced change detection algorithm by analyzing two images obtained at different times for the same region [3], [4], [5]. Accurate change detection is critical to disaster assessment [6], [7], agricultural evaluation, urban expansion [8], and so on, which bears great social and economic impacts.

In recent years, deep neural network (DNN)-based methods have achieved good performance in change detection, owing to their powerful representation learning capabilities. Depending on whether hand-marked labels are needed, the methods for change detection based on DNNs can be divided into supervised and unsupervised methods.

Supervised methods, which show good performance, rely on many labels for training. However, the annotation of aerial remote-sensing image data is usually expensive and unrealistic in practice. On the other hand, unsupervised methods that can generate change detection results without marked labels have been widely used in practice [5], [9], [10], [11]. These methods use traditional methods to generate change maps and use them as labels for training. The performance of the method depends on the quality of these labels. However, the labels generated by traditional change detection often do not have good quality. To reduce the influence of incorrect labels, patch-based methods and transfer-learning-based methods are proposed. Patch-based methods need to select labels with high confidence to train the model. Transfer-learning-based methods require obtaining extra knowledge from additional datasets for other tasks (such as image segmentation) and use transfer learning to guide the training [11], [12]. Patch-based and transfer-learning-based methods can alleviate the detection error caused by incorrect pixels to a certain extent. However, both types of methods have deficiencies: patch-based methods use small patches rather than the whole image as the training samples and classify whether the center pixel of an image patch changes. Such methods ignore the spatial information in the images; in addition, the small size of patch samples limits the variety of the network design. For transfer-learning-based methods, knowledge extracted from additional datasets may introduce new errors.

To rectify these problems, we propose an unsupervised progressive learning framework (UPLF) for optical aerial image change detection. We take the entire map as the training sample to avoid the problem of using small patches and propose an iterative and progressive method to further enhance the change detection performance without involving external knowledge. Empirically, the method can be largely converge to improved performance.

The novel contribution of this study embraces the following three aspects.

1) A new UPLF is proposed to iteratively enhance the change detection performance. Such a progressive update with label propagation can extend the reliable area from the initial change map while involving no external knowledge.
2) Our method takes the entire change map rather than the small patches as the training sample. To avoid incorrect labels in the entire map, a novel seed selection filter is

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designed to select the reliable regions in the estimated maps. By means of the seed selection filter, only those regions with strong confidence (termed seed thereafter) are deemed to be helpful in training.

3) An encode–decode network, based on the attention mechanism, is developed as the backbone of the progressive training to learn the representation of the differences between the optical aerial image pair.

This article is organized as follows. Section II reviews the related work. Section III describes the details of the proposed algorithm. In Section IV, we analyze the performance of the proposed method. Section V discusses the influence of related factors on the change detection results.

II. Related Work

Over the past few decades, a variety of change detection techniques have emerged, driven by great social and economic benefits. In most studies, two optical aerial images are generally compared to obtain a difference image (DI) [13]. Then, the DI is usually segmented using threshold and clustering methods [14], [15], [16], [17].

A. Traditional Methods

To generate a DI, many traditional methods have been widely used. Image differencing [18] and image ratioing [19] are straightforward techniques that perform subtraction and division operations, respectively, on the pixels of multitemporal images. Change-vector analysis (CVA) [20] compares each pair of corresponding pixels from the two optical multichannel images to obtain a change vector whose magnitude and direction represent the intensity and direction of the change. Principal component analysis (PCA) [21] is also used for change detection. It transforms high-dimensional images into low-dimensional feature spaces while retaining key information to reduce redundancy during the generation of the DI. Multivariate alteration detection (MAD) [22] attempts to maximize the variance of independently transformed variables and is invariant to linear scaling of the input data. With iteratively reweighed (IR) MAD [23], large weights are assigned to observations that show little change; observations that show large changes will have little weight.

B. Deep-Learning-Based Methods

In recent years, DNNs have become a popular research topic. A variety of DNNs have achieved good performance in change detection, such as deep belief networks, sparse autoencoders (AEs), recurrent neural networks (RNNs), and convolutional neural networks (CNNs). Differences between the optical aerial images can be learned effectively with the DNNs. The methods for change detection based on DNNs can be divided into supervised and unsupervised methods.

1) Supervised Methods: For remote-sensing images, supervised change detection methods have a considerable development [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35]. Besides, an amount of method are proposed for aerial images [2], [36], [37], [38], [39], [40], [41], [42], [43]. For example, in [39], a deep Siamese CNN with weighted contrastive loss [44] was proposed for aerial remote-sensing image change detection. In [40], an improved triplet loss function [41] was used to train a deep Siamese semantic network. In [42], three types of fully CNNs were proposed to perform change detection on two aerial remote-sensing images. In [43], a novel recurrent CNN that combines RNNs and CNNs was proposed to learn a spectral–spatial–temporal feature representation for change detection in multitemporal images.

The supervised methods mentioned above require an amount of hand-marked labels to train the models. However, the annotation of aerial remote-sensing image data is usually expensive and unrealistic in practice.

2) Unsupervised Methods: Unsupervised methods that can generate change detection results without marked labels have been widely used in practice for remote-sensing images [5], [9], [10], [11], [45], [46], [47], [48], [49], [50], [51], [52], [53], [54], [55]. Among them, the unsupervised change detection method for aerial images has made great progress.

In [55], an unsupervised change detection method based on difference representation learning is proposed. In [5], a symmetric CNN (SCCN) was used to transform input images into a consistent feature space, where DIs could be obtained. The parameters of the network were optimized using a coupling function without hand-marked labels. In [9], a PCANet with PCA filters was proposed for change detection. The interested pixels that have a high probability of being changed or unchanged are selected from the change maps obtained by traditional methods as labels, and the small patches around the pixels are cropped from the image as the training samples. After the training, the trained neural network is used to classify the remaining pixels in the image. In [10], an unsupervised change detection method was proposed based on generative adversarial networks (GANs). It uses typical methods to generate labels for GAN training. In [11], a CNN based on transfer learning (TLCNN) for change detection in optical aerial images was trained on an open dataset for supervised semantic segmentation and labels generated by the log-ratio algorithm. In [56], an improved GAN is proposed for unsupervised change detection. In [12], transfer learning is effectively used in change detection. Small patches are cropped from original images to be as training samples.

In this article, we proposed an unsupervised change detection method for aerial images.

III. Methodology

UPLF uses a progressive DNN for change detection in optical aerial images. As shown in Fig. 1, the proposed method consists of two main stages: seed selection filtering and progressive unsupervised learning.

In the first stage, a typical change detection method is applied to generate rough original change maps; then, a seed selection filter is designed to generate a seed-selection map, which makes sure that only pixels with reliable labels are seen as seeds to guide the training.

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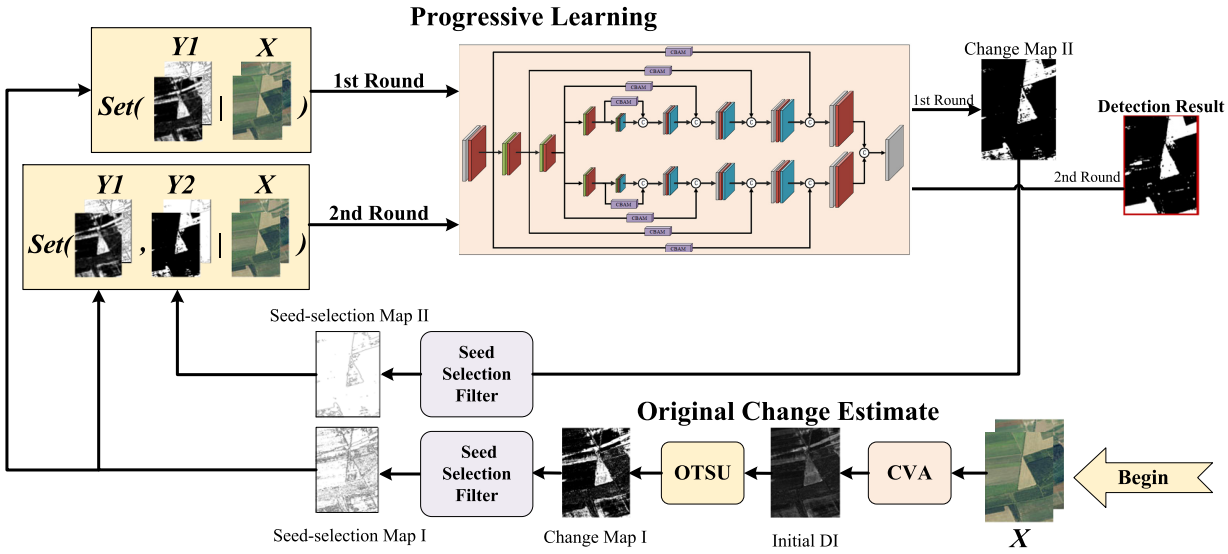


Fig. 1. UPLF: 1) In the original change estimation stage, original estimated change maps are generated as labeled samples, and the seed regions are chosen through the seed selection filter. 2) In the progressive learning stage, the designed network is first trained on the seed regions; then, the output of the network is used to extend the seeds. The network is progressively trained to generate the detection result.

In the second stage, progressive unsupervised learning consists of two rounds of training, and more change maps are generated progressively. In the first round of training, we use the original change and seed-selection maps as labels to train the network. Parts of the pixels are labeled while others are unlabeled. As the output of the network, another set of change maps is generated. In the second round of training, we once again used the seed selection filter to generate seed-selection maps. We combined the original and the newly generated seed-selection maps to extend the seed area. Then, the extended seeds are used to retrain the network and generate the final change detection result.

A. Seed Selection Filter

1) Original Change Estimation: Two coregistered intensity remote-sensing images $\mathbf{X}_1 = \{x_1(i, j), 1 \le i \le H, 1 \le j \le W\}$ and $\mathbf{X}_2 = \{x_2(i, j), 1 \le i \le H, 1 \le j \le W\}$, have the same size, $H \times W$, and are acquired at the same geographical area at different times, t_1 and t_2, respectively. (i, j) represents the pixel coordinates in row i and column j of the image. We utilize CVA to deal with the two images to generate the DI, where $\mathbf{DI}_1 = \{\mathrm{di}_1(i, j), 1 \le i \le H, 1 \le j \le W\}$. Then, we use Otsu [57] to classify the pixels of $\mathbf{DI}_1$ into two different classes, namely, changed and unchanged classes; thus, Change Map I $\mathbf{CM}_1$ can be obtained. As a thresholding algorithm, Otsu can automatically determine the threshold, thus providing an advanced and easy implementation to generate the original change detection map.

2) Principle of the Filter: The original change map generally contains some pixels that are incorrectly classified. Both correct and erroneous pixels are used as training samples. Thus, there is a high chance that erroneous pixels will mislead network training. Existing methods crop reliable small patches from the original change maps as training samples [9], [10], [56], however, the use of small patches ignores the spatial information of the whole images. To address this issue, we provide an image filter to select seeds to guide the training.

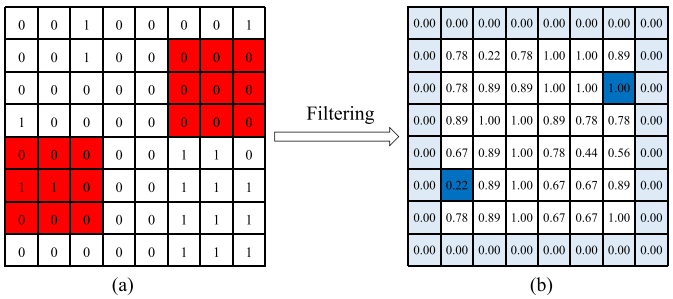


Fig. 2. Intuitive exhibition of the filtering process. The filtering result of the red region in the change map corresponds to the dark blue in the filtered map. A zero is padded along the edges of the filtered map to ensure the pixels of the image before and after filtering remain in a one-to-one correspondence. (a) Change map. (b) Seed-selection map.

Fig. 2 shows an intuitive representation of the filtering process. Fig. 2(a) is an example of a change map. Let $\mathrm{cm}_k(i, j)$ represent a pixel in a change map, $\mathbf{CM}_k = \{\mathrm{cm}_k(i, j), 1 \le i \le H, 1 \le j \le W\}$, and let k be used to distinguish it. Fig. 2(b) is the filtered result, which is denoted as the seed-selection map. Let $\mathrm{sm}_k(i, j)$ represent the value of pixel (i, j) in the seed-selection map.

As shown in Fig. 2(a), the pixel $\mathrm{cm}_k(3, 7)$ (the third row and seventh column) has a neighborhood of size 3×3, which is highlighted in red. All pixels in this neighborhood area have the same label; in this case, it has a high probability that $\mathrm{cm}_k(3, 7)$ is a seed in the change map. Meanwhile, the label of pixel $\mathrm{cm}_k(6, 2)$ differs from its adjacent labels; thus, the label of $\mathrm{cm}_k(6, 2)$ has a low probability of being seed.

We define a patch having a center of $\mathrm{cm}_k(i, j)$ and size $w \times w$ (w is odd). Then, $\mathrm{cm}_k(i + l, j + l)$ represents pixels in the patch, and l is the offset of the position coordinates, $-(1/2)(w - 1) \le l \le (1/2)(w - 1)$. The patch is filtered as follows:

$$\mathrm{sm}_k(i, j) = \sum_{l=-\frac{1}{2}(w-1)}^{\frac{1}{2}(w-1)} \mathrm{cm}_k(i + l, j + l) \odot \frac{\mathrm{cm}_k(i, j)}{w \times w}. \quad (1)$$

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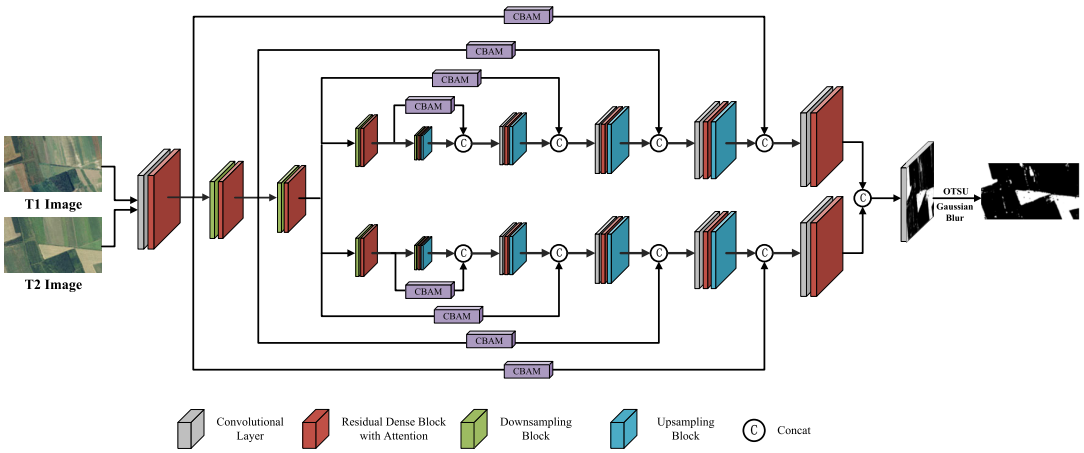


Fig. 3. Backbone network of the progressive training.

Here, $\odot$ represents exclusive the XNOR, which is a mathematical operator applied to logical operations. If $\mathrm{cm}_k(i, j)$ and $\mathrm{cm}_k(i+l, j+l)$ have the same label, the result of $\odot$ is one. If they have different labels, the result of $\odot$ is zero. According to (1), $\mathrm{sm}_k(i, j)$ is the ratio of the number of pixels having the same values as the center pixel to the total number of pixels in the patch. Therefore, $\mathrm{sm}_k(i, j)$ can be used to represent the probability that the pixel $\mathrm{cm}_k(i, j)$ is the seed.

For pixels at the edges of the original change map, we cannot find a neighborhood surrounding them. Therefore, filtering cannot be carried out on the edge pixels, and the size of the map after filtering is reduced. To ensure that the size of the map after filtering is the same as before, we pad zeroes to the edges of the filtered map, as shown in Fig. 2. In practice, the change maps obtained generally have large sizes; therefore, the padding of pixels hardly affects subsequent operations.

Through filtering, the unreliable pixels tend to have a lower value in the filtered map than do the seed pixels which will guide the training. Our method can cope with large images with the help of a seed selection filter. Thus, the spatial features of the corresponding input aerial images are well-preserved, and the representation learning ability of the networks can be fully utilized.

B. Progressive Learning

The proposed progressive learning framework consists of two rounds of the iterative training process, as depicted in Fig. 1. The training is conducted on a backbone network.

1) Backbone Network: We propose an encode–decode network based on the attention mechanism to learn the representation of the differences between the optical aerial image pair. It is used as the backbone network of the progressive learning framework.

As shown in Fig. 3, we use the shallow layers in the feature extraction module to simultaneously extract low-level features of the two images. Then, the high-level semantic features of the two images are separately extracted and decoded by the two branches having the same structure but different parameters. A residual dense block with attention mechanism [58] is used as the basic block of the network. To obtain more context information, the downsampling operation is used in the encoder. However, the downsampling operation often results in the loss of detailed information about the feature. We utilize the skip connection to combine the features of the encoder with that of the decoder. Besides, we use a convolutional block attention module (CBAM) [59] to highlight the useful feature while suppressing the useless feature for the decoder.

The DI is reconstructed to the same size as the original image. The change detection results can be obtained using a threshold method (such as Otsu) on the DI.

2) First Round Training: Given two optical aerial remote-sensing images $\mathbf{X}_1$ and $\mathbf{X}_2$, $\mathbf{DI}_2 = \{\mathrm{di}_2(i, j), 1 \le i \le H, 1 \le j \le W\}$ is the output of the network, which can be defined as follows:

$$\mathbf{DI}_2 = f_{\theta_r}\left(g_{\theta_e}\left(\mathbf{X}_1, \mathbf{X}_2\right)\right) \tag{2}$$

where g and θ_e correspond to the encode module of the network and its parameters, respectively. f and θ_r correspond to the decode module of the network and its parameters, respectively.

Change detection is treated as a binary classification problem; therefore, we use the binary cross-entropy loss function, l_{bce}, which is defined as follows:

$$l_{\mathrm{bce}}(y_1, y_2) = -y_1 \times \log(y_2) - (1 - y_1) \times \log(1 - y_2) \tag{3}$$

where y_1 and y_2 are the parameters in the function.

Given the Change Map I $\mathbf{CM}_1$ and the Seed-selection Map I $\mathbf{SM}_1$ generated in the original change estimation stage, for the pixels located at (i, j) in the kth pair of images of a training batch having a size of N, the loss function of the network is defined as

$$L^{1\mathrm{st}} = \sum_{k=1}^{N}\sum_{i=1}^{H}\sum_{j=1}^{W} \mathrm{sm}_1^*(i, j) \times l_{\mathrm{bce}}\left(\mathrm{cm}_1(i, j), \mathrm{di}_2(i, j)\right) \tag{4}$$

where $\mathrm{sm}_1^*(i, j)$ is defined as follows:

$$\mathrm{sm}_1^*(i, j) = \begin{cases} \mathrm{sm}_1(i, j), & \mathrm{sm}_1(i, j) \ge \alpha \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

Here, α is a threshold. $\mathrm{sm}_1^*(i, j)$ is set to 0 if $\mathrm{sm}_1(i, j)$ is below α; thus, only the seed are can guide the training.

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After the network is trained by the first round, the original bitemporal aerial images are fed into the network. And the change map $\mathbf{CM}_2 = \{\mathrm{cm}_2(i, j), 1 \le i \le H, 1 \le j \le W\}$ can be obtained as another set of labels.

3) Second Round Training: We still use the designed filter to deal with the change map $\mathbf{CM}_2$. Then the Seed-selection Map II $\mathbf{SM}_2 = \{\mathrm{sm}_2(i, j), 0 \le i \le H, 1 \le j \le W\}$, where $0 \le \mathrm{sm}_2(i, j) \le 1$, can be obtained. $\mathbf{SM}_2$ will be used to control the usage of seed area in $\mathbf{CM}_2$ during the progressive process.

During the progressive training stage, Change Map I $\mathbf{CM}_1$, Seed-selection Map I $\mathbf{SM}_1$, the Change Map II $\mathbf{CM}_2$, and Seed-selection Map II $\mathbf{SM}_2$ are jointly used to further train the UPLF. With the progressive process, the seed area has been expanded. Thus, the network is further updated and can generate a more accurate change map.

Given two optical aerial remote-sensing images, $\mathbf{X}_1$ and $\mathbf{X}_2$, the output of the network is the DI, $\mathbf{DI}_f = \{\mathrm{di}_f(i, j), 1 \le i \le H, 1 \le j \le W\}$, which can be defined as follows:

$$\mathbf{DI}_f = \varphi_{\eta_r}\left(\xi_{\eta_e}(\mathbf{X}1, \mathbf{X}2)\right) \tag{6}$$

where ξ and η_e correspond to the encode module of the network and its parameters, respectively. φ and η_r correspond to the decode module of the network and its parameters, respectively.

The loss function to the progressive optimization is defined as

$$L^{2\mathrm{nd}} = \beta L_1^{2\mathrm{nd}} + (1 - \beta) L_2^{2\mathrm{nd}}. \tag{7}$$

Here, $L_1^{2\mathrm{nd}}$ is the loss function based on $\mathbf{CM}_1$, and $L_2^{2\mathrm{nd}}$ is the loss function based on $\mathbf{CM}_2$. $0 \le \beta \le 1$ is a constant parameter that is used to balance the effects of $\mathbf{CM}_1$ and $\mathbf{CM}_1$.

$$L_1^{2\mathrm{nd}} = \sum_{k=1}^{N}\sum_{i=1}^{H}\sum_{j=1}^{W} \mathrm{sm}_1^*(i, j) \times l_{\mathrm{bce}}\left(\mathrm{cm}_1(i, j), \mathrm{di}_f(i, j)\right) \tag{8}$$

$$L_2^{2\mathrm{nd}} = \sum_{k=1}^{N}\sum_{i=1}^{H}\sum_{j=1}^{W} \mathrm{sm}_2^*(i, j) \times l_{\mathrm{bce}}\left(\mathrm{cm}_2(i, j), \mathrm{di}_f(i, j)\right) \tag{9}$$

where $\mathrm{sm}_2^*(i, j)$ is obtained using the same strategy as $\mathrm{sm}_1^*(i, j)$

$$\mathrm{sm}_2^*(i, j) = \begin{cases} \mathrm{sm}_2(i, j), & \mathrm{sm}_2(i, j) \ge \alpha \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

IV. Experiments

To validate the effectiveness of the proposed method, we tested the change-detection problem in optical aerial images on two real datasets. Considering that our method is unsupervised, we selected a series of unsupervised change detection methods as comparison algorithms, including image ratioing [19], CVA [20], PCA [21], MAD [22], IR-MAD [23], S3VM [60], GAN [10], and TLCNN [11]. Besides, we also compared our method with some supervised methods, such as CNN [38], SCCN [39], fully convolutional early fusion (FC-EF) [42], Fully Convolutional Siamese Concatenation (FC-Siam-conc) [42], and deep depthwise separable convolutional network (DDSCN) [38].

A. Datasets

Our experiments are based on the SZTAKI AirChange Benchmark set which has been widely used in aerial remote-sensing image change detection. This dataset was provided by the Hungarian Institute of Geodesy Cartography and Remote Sensing (FöMI) and Google Earth [61]. The dataset consists of three parts (i.e., SZADA, TISZADOB, and ARCHIEVE), containing seven, five, and one subdatasets, respectively. The labels for each dataset were obtained via expert labeling. Each subdataset of SZADA contains two aerial remote-sensing images which are, respectively, obtained in 2000 and 2005. The size of the images was 952 × 640 pixels. Each subdataset of TISZADOB contains a pair of 952 × 640 images obtained in 2000 and 2007. The ARCHIVE dataset contains a pair of 952 × 640 images, respectively, obtained in 1984 and 2007. The quality of the aerial images taken in 1984 was poor, and therefore, the ARCHIVE dataset was not used in our experiments. Additionally, to meet the requirements of unsupervised tasks, only the reference labels in the dataset were used in the evaluation.

According to other change detection methods [11], [39], [40], [42], we chose subdatasets SZADA/1 and TISZADOB/3 to evaluate the performance of the proposed method. The left corner having a size of 784 × 448 pixels of aerial images was used as the test data, as shown in Fig. 4, referring to the processing used in [11], [39], and [40]. The rest of the region was cropped to 112 × 112 in an overlapped fashion to generate the training data. To augment the training data, we rotated the cropped images by 90°, 180°, and 270° and flipped them horizontally and vertically. Then we obtained 3744 training image pairs.

B. Evaluation Measures and Experimental Settings

We used three measures [i.e., precision (Pr), recall (Rc), and F1-measure (F1)] to evaluate the performance of the proposed method.

Precision represents the percentage of the pixels classified correctly into the changed class among all pixels classified into the changed class. This can be defined as follows:

$$\mathrm{Pr} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{11}$$

where TP and FP represent the number of pixels classified into the changed class correctly and the number of pixels classified into the changed class incorrectly.

Recall represents the percentage of the pixels classified into the changed class correctly among all actual changed pixels. This can be defined as follows:

$$\mathrm{Rc} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{12}$$

where FN represents the number of pixels classified into the unchanged class wrongly.

The F-measure is the harmonic mean of precision and recall. This is calculated as follows:

$$\mathrm{F} = \frac{\left(a^2 + 1\right) \cdot \mathrm{Pr} \cdot \mathrm{Rc}}{a^2 \cdot (\mathrm{Pr} + \mathrm{Rc})}. \tag{13}$$

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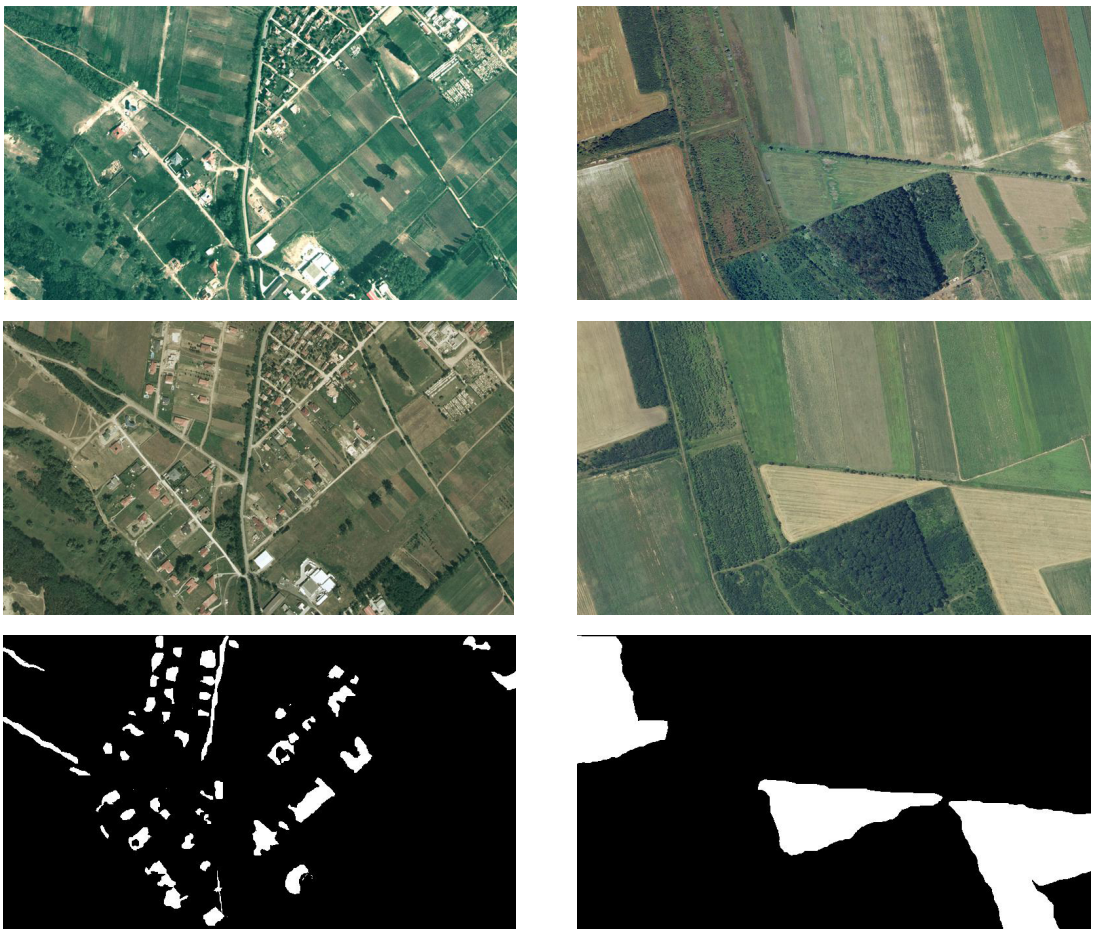


Fig. 4. (a) SZADA/1 and (b) TISZADOB/3 datasets. Changes are displayed in white regions.

TABLE I
CHANGE DETECTION RESULT OF THE UNSUPERVISED METHOD ON THE SZADA/1 DATASET

Method	Pr(%)	Rc(%)	F1(%)
ImageRatio	11.9	56.3	19.7
CVA	20.7	47.8	28.9
PCA	18.3	61.1	28.1
MAD	17.1	**66.8**	27.2
IRMAD	21.5	52.3	30.5
S3VM	19.2	48.7	27.5
GAN	28.3	40.9	33.5
TLCNN	27.2	56.1	36.7
Ours	**41.3**	45.7	**43.4**

When a is set to 1, the F1-measure can be obtained as

$$\mathrm{F1} = \frac{2 \cdot \mathrm{Pr} \cdot \mathrm{Rc}}{\mathrm{Pr} + \mathrm{Rc}}. \tag{14}$$

The networks were trained using Pytorch on an NVIDIA GeForce GTX TITAN with 12-GB GPU. Adam [62] was used as the optimizer used to train the network. We used the Xavier initializer [63] to initialize the weights of each layer of the network. The initial learning rate was set to 0.0001.

TABLE II
CHANGE DETECTION RESULT OF THE SUPERVISED METHOD ON THE SZADA/1 DATASET

Method	Pr(%)	Rc(%)	F1(%)
CNN	47.1	53.4	50.1
SCNN	38.2	46.6	42.0
FC-EF	39.1	48.2	43.2
FC-Siam-conc	43.0	45.8	44.3
DDSCN	48.2	56.1	51.8
Ours	41.3	45.7	43.4

C. Comparison on the SZADA/1 Dataset

We compared our method with other state-of-the-art change detection methods for optical aerial images. To make our comparison more convincing, we select unsupervised methods with public code in recent years as the comparison algorithms. Moreover, we also compare with some supervised methods to further demonstrate the effectiveness of the proposed method. Fig. 5 shows the change maps obtained by the various methods on the SZADA/1 dataset. The dataset contains many small changed areas that make change detection difficult. Table I lists the results of the quantitative comparison of unsupervised methods. The proposed method exhibited the best performance

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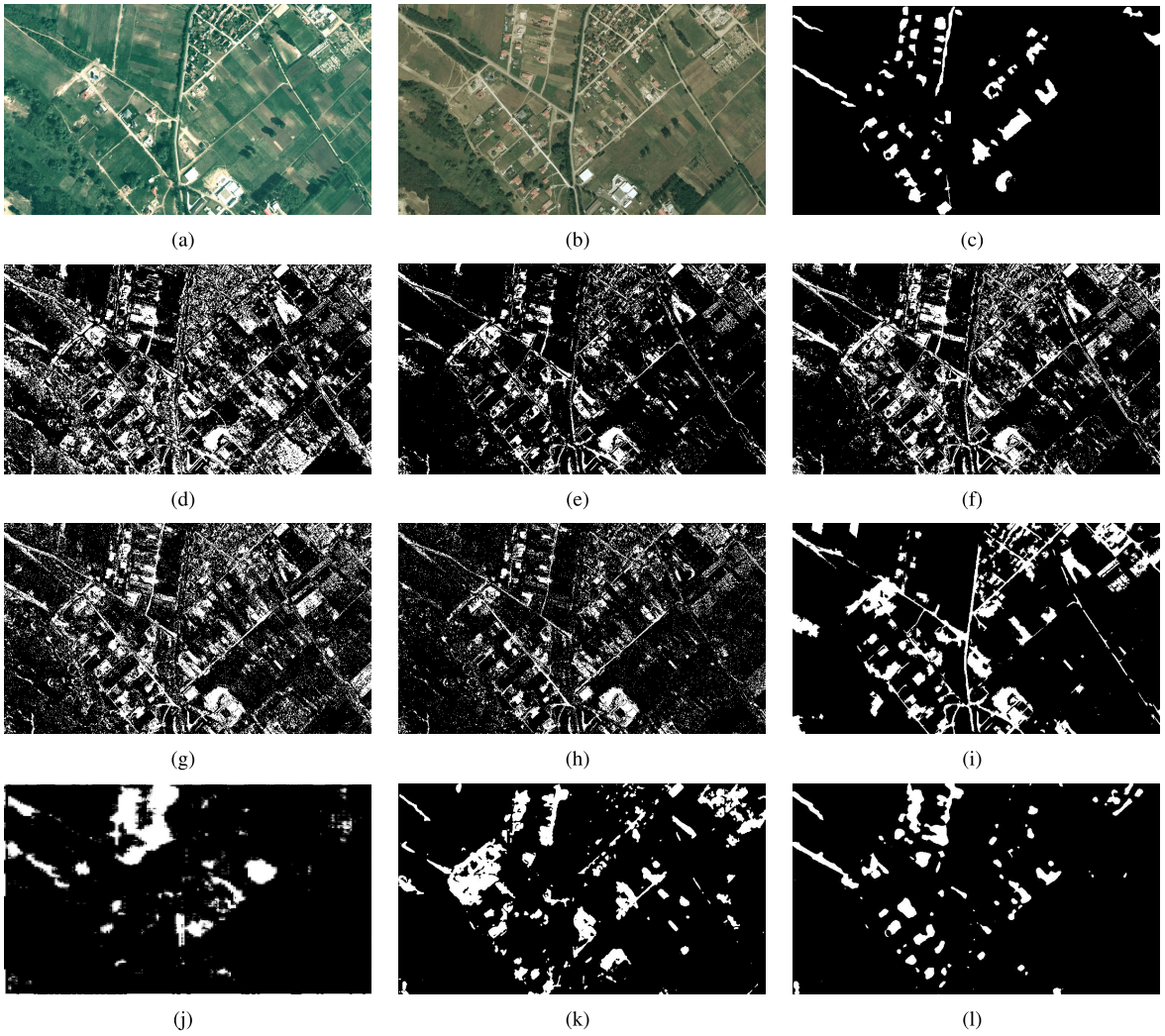


Fig. 5. Change maps obtained by different methods on the SZADA/1 dataset: (a)–(c) Original images and reference image; (d) ImageRatio; (e) CVA; (f) PCA; (g) MAD; (h) IRMAD; (i) S3VM; (j) GAN; (k) TLCNN; and (l) UPLF.

and significantly outperformed the second best. Comparison results with supervised change detection methods are shown in Table II. Our method has better performance than FC-EC. These show that our unsupervised change detection method can reach or exceed some supervised change detection methods, which further confirms the superiority of our method.

Fig. 5(a)–(e) shows the change detection results obtained by traditional methods, including ImageRatio, CVA, PCA, MAD, and IRMAD. As can be seen, the change maps obtained by traditional methods contain many white-noisy points. ImageRatio yielded the noisiest points, as shown in Fig. 5(a), which led to the lowest precision. We can hardly distinguish the main changed areas. The results of CVA and PCA contain fewer noisy points than does ImageRatio at the bottom-left and bottom-right corners of Fig. 5(b) and (c). MAD had the highest recall, as listed in Table I, but the precision and F1-measure were low, because most pixels classified as changed were the wrong noisy points, as shown in Fig. 5(d). As the MAD was extended, IRMAD generated fewer noisy points and gained a higher F1-measure than MAD did, as shown in Fig. 5(e).

Fig. 5(f)–(h) shows the change maps obtained by existing training-based methods, including S3VM, GAN, and TLCNN. Compared with the traditional methods described above, the change maps obtained by the training-based methods contained almost no white-noisy points. However, these results remain problematic. As shown in Fig. 5(f), many small changed areas are undetected, and many unchanged areas are detected as changed, which is particularly serious at the upper-right corner of the image. As shown in Fig. 5(g), many small areas were misdetected, resulting in the main change areas being barely visible. As shown in Fig. 5(h), the combination of transfer learning and CNNs caused the TLCNN method to yield a change map having fewer errors, but there are still some unchanged areas detected as changed, such as the one at the upper-right corner of the image.

The change map obtained by the proposed method is shown in Fig. 5(i). We can see that the problem of misdetection was significantly alleviated. From Table I, we can see that our proposed method has the highest precision. Furthermore, precision and recall reached a good balance, and the highest F1-measure was obtained.

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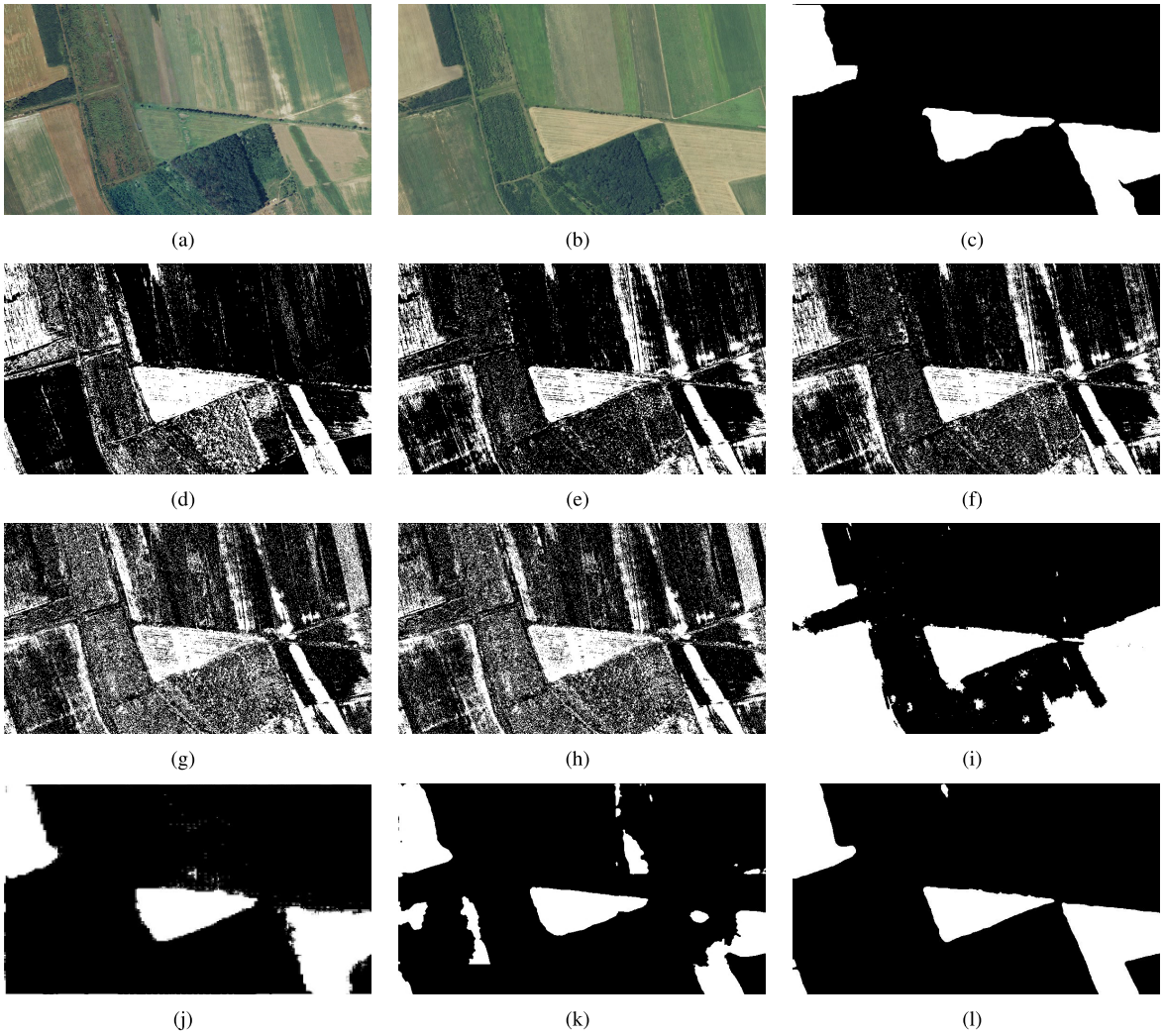


Fig. 6. Change maps obtained by different methods on the TISZADOB/3 dataset: (a)–(c) Original images and reference image; (d) ImageRatio; (e) CVA; (f) PCA; (g) MAD; (h) IRMAD; (i) S3VM; (j) GAN; (k) TLCNN; and (l) UPLF.

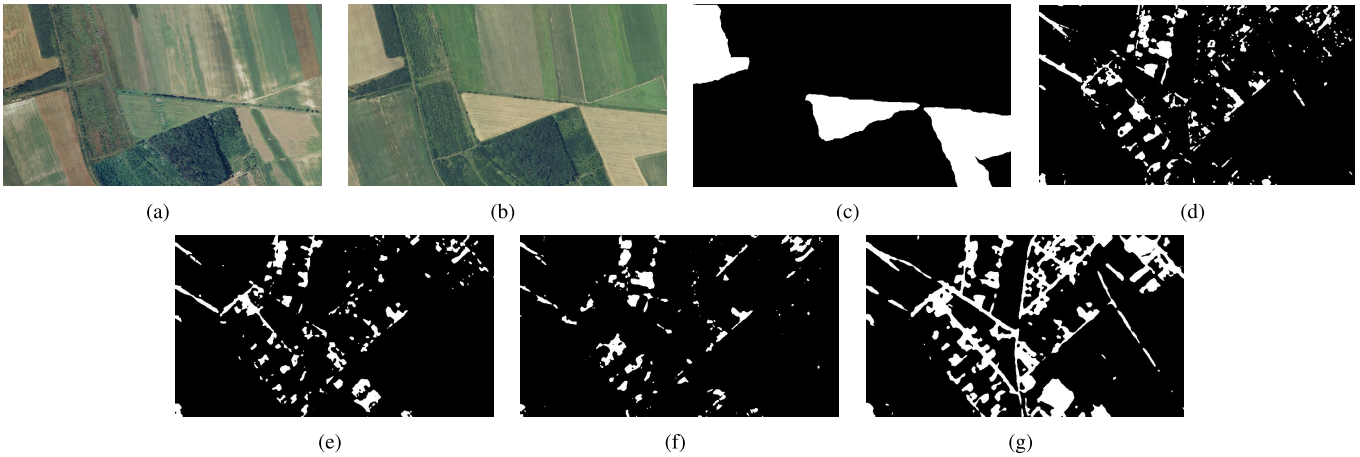


Fig. 7. Change maps obtained with different neighborhood sizes in the filtering on the SZADA/1 dataset: (a)–(c) Original images and reference image; (d) $w = 3$; (e) $w = 5$; (f) $w = 7$; and (g) $w = 9$.

D. Comparison on the TISZADOB/3 Dataset

We show the hand-marked label and change maps obtained by the various methods on the TISZADOB/3 dataset in Fig. 6. Table III lists the results of the quantitative comparison of unsupervised methods. As we can see, the proposed method exhibited the best performance and significantly outperformed the second best. Comparison results with supervised change detection methods are shown in Table IV. Our method has the best performance among the compared methods.

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TABLE III
CHANGE DETECTION RESULT OF EACH METHOD ON THE TISZADOB/3 DATASET

Method	Pr(%)	Rc(%)	F1(%)
ImageRatio	46.3	67.4	57.9
CVA	37.6	54.0	44.3
PCA	36.4	57.4	44.6
MAD	22.8	47.5	30.8
IRMAD	25.3	50.9	33.8
S3VM	43.3	91.1	58.7
TLCNN	78.7	71.7	75.0
GAN	87.9	92.5	90.1
Ours	**94.0**	**96.9**	**95.4**

TABLE IV
CHANGE DETECTION RESULT OF THE SUPERVISED METHOD ON THE TISZADOB/3 DATASET

Method	Pr(%)	Rc(%)	F1(%)
CNN	83.4	97.4	89.8
SCNN	93.4	91.5	92.5
FC-EF	97.1	81.5	88.6
FC-Siam-conc	95.5	94.2	94.9
DDSCN	93.4	91.5	92.4
Ours	94.0	96.9	95.4

The results of the traditional methods (i.e., ImageRatio, CVA, PCA, MAD, and IRMAD) are shown in Fig. 6(a)–(e). Many white-noisy points can be observed in the change maps. ImageRatio failed to detect some main changed areas at the bottom-right corner of the image in Fig. 6(a). CVA and PCA detected some unchanged areas as changed at the upper-right and bottom-left corners of the image, as shown in Fig. 6(b) and (c). MAD and IRMAD did not perform well on this dataset, and their results contained too many noisy points, as shown in Fig. 6(d) and (e).

Compared with the traditional methods mentioned above, the change maps obtained by the training-based methods contained almost no white-noisy points, as shown in Fig. 6(f)–(h). From the result of S3VM, as shown in Fig. 6(f), most changed areas could be detected with few noisy points, leading to a high recall. However, there was a large unchanged area detected as changed at the bottom-left corner of the image. The problem of undetected and detected-by-mistake areas was alleviated by the TLCNN-based transfer learning, as shown in Fig. 6(g). The GAN-based method has a better performance than the above methods, as shown in Fig. 6(h).

Fig. 6(i) shows the change map obtained by the proposed method. Nearly all changed areas can be detected. Additionally, there were far fewer areas that were mistakenly changed.

Quantitative comparison results with unsupervised methods are presented in Table III. There, we see that our method was the best with the evaluation of all three measures. It is obvious that our proposed method has a better performance than the others from qualitative and quantitative analyses.

TABLE V
CHANGE DETECTION COMPARISON WITH DIFFERENT NEIGHBORHOOD SIZE

Dataset	Metrics	Size of neighborhood			
		3	5	7	9
SZADA/1	Pr(%)	32.73	**35.18**	34.03	18.71
	Rc(%)	46.14	42.59	30.10	**59.68**
	F1(%)	38.29	**38.53**	31.95	28.49
TISZADOB/3	Pr(%)	93.49	93.73	**93.95**	91.34
	Rc(%)	88.83	91.76	83.04	**96.44**
	F1(%)	91.10	**92.73**	88.16	93.82

E. Complexity Analysis

In this section, we analyze the complexity of the proposed method. The time complexity of a single training time is $O(n)$. In our method, we need to train the backbone network multiple times, and the time complexity is $O(kn)$, where k is the training times. The time complexity has not been boosted exponentially. To sum up, our method greatly improves the performance while only moderately increasing the time complexity.

V. DISCUSSION

During the learning process of the network, the related factors influenced the performance of the algorithm. We next analyze the effects of these factors on the performance of the proposed method.

A. Effects of the Neighborhood Size

During image filtering, neighborhood size w in (1) determines the area on which the filter focused. If w is set to be smaller, the pixels in the small noise areas may be considered as correctly labeled. If w is set to be larger, the neighborhood of the pixel will contain redundant information, and the neighborhood filtering of the correct label pixels may be affected by the surrounding noise pixels. In the experiment, we set w to 3, 5, 7, and 9 to investigate the influence of the neighborhood size on the change detection result. With different neighborhood sizes, the change maps and evaluation indexes are presented in Figs. 7 and 8 and Table V. When w was set to 5, the F1-measure as a harmonic measure was high on both datasets; moreover, the precision achieved the highest value on the SZADA/1 dataset. When w was set to 7, precision on the TISZADOB/3 dataset achieved the highest value; however, the performance on other metrics was mediocre. The performance of the proposed method is worse when w is set to 9, because the neighborhood contained too much redundant information. Therefore, it is appropriate that w is set to 5.

B. Effects of Parameter β

Parameter β in the loss function (7) determines the influences of the change maps generated by the traditional methods and by the first trained network. To study the effects of β, we changed its value and obtained the corresponding change

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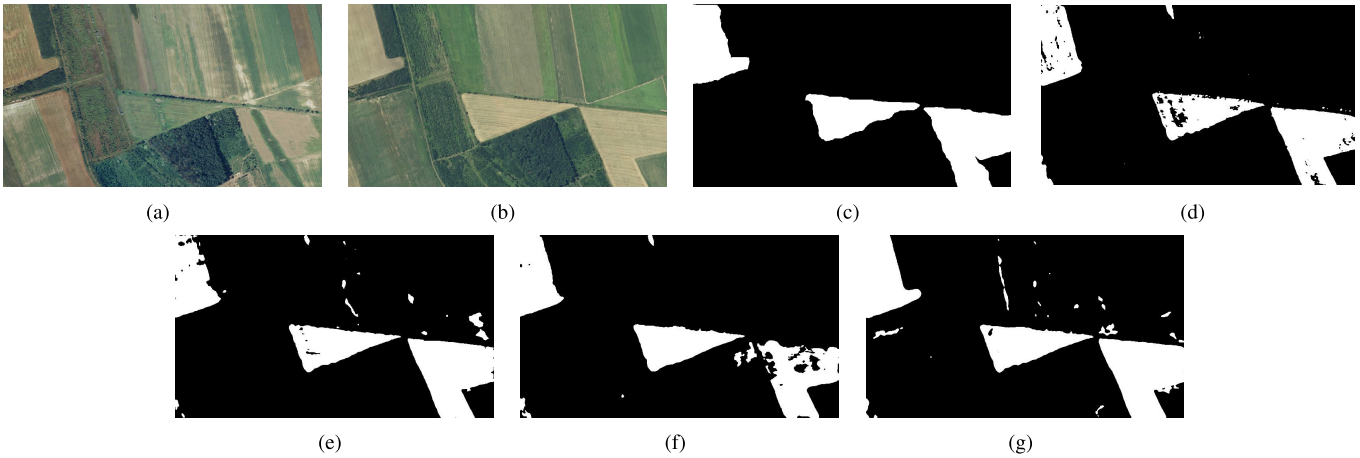


Fig. 8. Change maps obtained with different neighborhood sizes in the filtering on the TISZADOB/3 dataset: (a)–(c) Original images and reference image; (d) $w = 3$; (e) $w = 5$; (f) $w = 7$; and (g) $w = 9$.

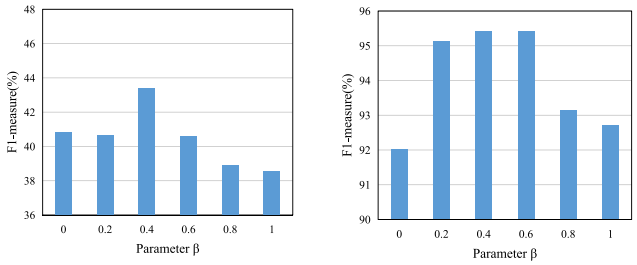


Fig. 9. Relationship of F1-measure and the parameter β on the (a) SZADA/1 and (b) TISZADOB/3 datasets.

TABLE VI
CHANGE DETECTION RESULT OF WHETHER SEED SELECTION FILTERING IS USED IN THE FIRST TRAINING

Dataset	Metrics	Using filter or not	
		No	Yes
SZADA/1	Pr(%)	17.99	**35.18**
	Rc(%)	**63.26**	42.59
	F1(%)	28.02	**38.53**
TISZADOB/3	Pr(%)	84.88	**93.73**
	Rc(%)	**95.96**	91.76
	F1(%)	90.08	**92.73**

TABLE VII
CHANGE DETECTION RESULT OF WHETHER IMAGE FILTERING IS USED IN PROGRESSIVE TRAINING

Dataset	Metrics	Using filter or not	
		No	Yes
SZADA/1	Pr(%)	35.83	**41.28**
	Rc(%)	44.49	**45.65**
	F1(%)	39.69	**43.36**
TISZADOB/3	Pr(%)	91.39	**94.05**
	Rc(%)	94.52	**96.86**
	F1(%)	92.93	**95.43**

TABLE VIII
CHANGE DETECTION RESULT USING DIFFERENT TRAINING TIMES

Dataset	Metrics	Training times		
		First	Second	Third
SZADA/1	Pr(%)	35.18	41.28	**42.68**
	Rc(%)	42.59	**45.65**	41.41
	F1(%)	38.53	**43.36**	42.03
TISZADOB/3	Pr(%)	93.73	94.05	**95.50**
	Rc(%)	91.76	**96.86**	95.77
	F1(%)	92.73	95.43	**95.64**

detection results. We set β to be 0, 0.2, 0.4, 0.6, 0.8, and 1. In particular, when β was set to 0, only the change maps generated by the trained network participated in the progressive training. When β was set to 1, the change maps generated by the traditional methods participated in network training. We presented the relationship between F1-measure and the parameter β on the two datasets in Fig. 9. When the parameter β was set to 0.6, the proposed method exhibited the best performance on the two datasets.

C. Effects of the Filtering

The seed selection filter was designed to restrain the adverse impact of incorrect information in change maps.

Table VI presents the performance of the first trained network with and without the seed selection filter. On the two datasets, when the filter was not used, although recall was higher, precision and F1-measure were very low. This is because many unchanged pixels were incorrectly classified as changed. When the filter was used, the results obtained by the proposed method achieved a balance between precision and recall and obtained a higher F1-measure.

Table VII presents the performance of the progressive trained network with and without the seed-filter operation. When the filter is used, all the measures are higher on the two datasets.

Fig. 10 shows the change maps obtained by the initialized trained network with or without filtering. Fig. 11 shows the change maps obtained by the progressive trained network with or without filtering. We can find that when filtering is used, the change maps are closer to the ground truth.

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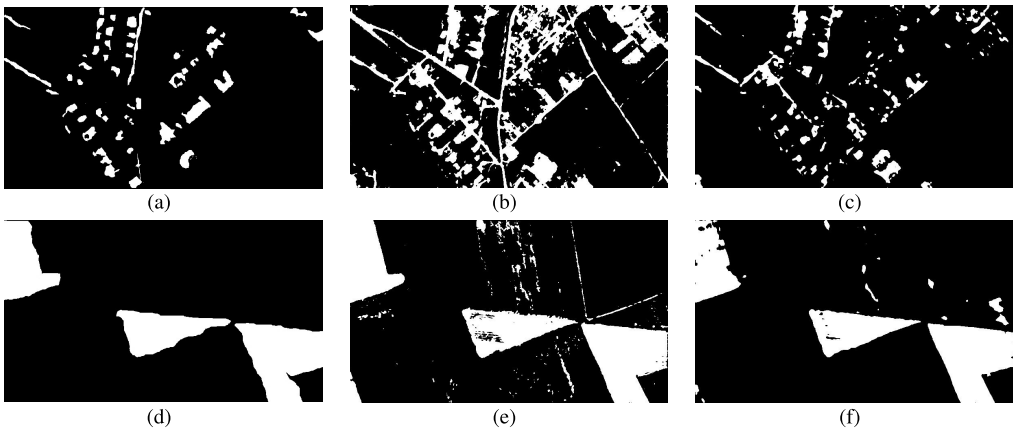


Fig. 10. Change maps obtained by the trained network with or no filtering. On the SZADA/1 dataset: (a) reference image; (b) no filter; and (c) with filter. On the TISZADOB/3 dataset: (d) reference image; (e) no filter; and (f) with filter.

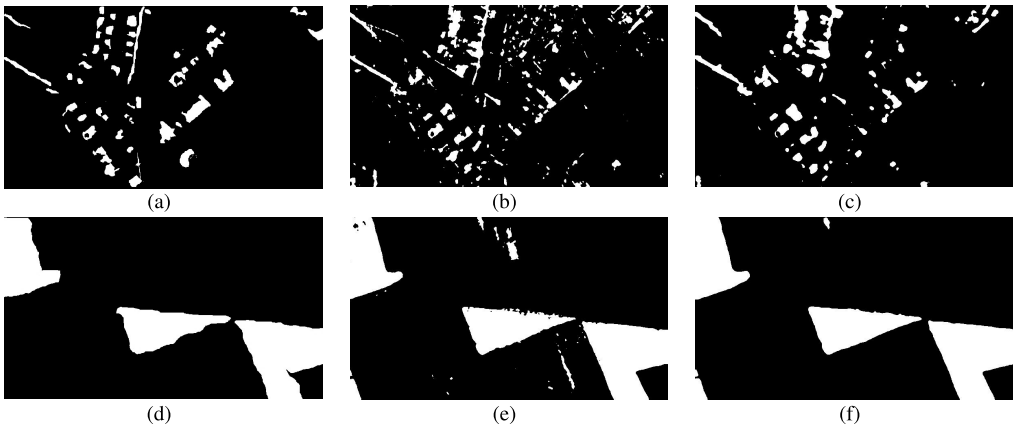


Fig. 11. Change maps obtained by the progressive trained network with or no filtering. On the SZADA/1 dataset: (a) reference image; (b) no filter; and (c) with filter. On the TISZADOB/3 dataset: (d) reference image; (e) no filter; and (f) with filter.

Generally, the use of the seed selection filter can effectively improve the effect of change detection.

D. Effects of the Progressive Training Times

To study the influence of the progressive training times, we performed an additional third progressive training in this section. The training labels are the change maps generated by the first and second rounds of training. The results of the experiment are shown in Table VIII. On the SZADA/1 dataset, the F1-score is declining in the third round, and on the TISZADOB/3 dataset, the F1-score is rising. By trading off the cumbersomeness of the method and the improvement of effect, we use the strategy of twice training network.

VI. Conclusion

In this article, we proposed a novel unsupervised change detection method, UPLF, for optical aerial images. We provided an unsupervised progressive training strategy that iteratively trained the network. We also proposed a seed selection filter method to suppress the impact of incorrect information in labels. We expand the seed area to progressively train the network, avoiding the problem of using small patches, while without involving external knowledge. The encode–decode network based on the attention mechanism is designed as the backbone of the progressive model to learn the representation of the differences between the paired optical aerial images. The proposed method is validated on two commonly used datasets SZADA/1 and TISZADOB/3. On both datasets, our method achieves F1 scores of 43.4% and 95.4%, with an average improvement of above 12% compared with the second-best method.

References

[1] F. Liu, L. Jiao, X. Tang, S. Yang, W. Ma, and B. Hou, "Local restricted convolutional neural network for change detection in polarimetric SAR images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 3, pp. 818–833, Mar. 2019.

[2] H. Zhang, M. Lin, G. Yang, and L. Zhang, "ESCNet: An end-to-end superpixel-enhanced change detection network for very-high-resolution remote sensing images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 1, pp. 1–15, Jan. 2021.

[3] E. M. Domínguez, E. Meier, D. Small, M. E. Schaepman, L. Bruzzone, and D. Henke, "A multisquint framework for change detection in high-resolution multitemporal SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3611–3623, Jun. 2018.

[4] M. Gong, J. Zhao, J. Liu, Q. Miao, and L. Jiao, "Change detection in synthetic aperture radar images based on deep neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 1, pp. 125–138, Jan. 2016.

[5] J. Liu, M. Gong, K. Qin, and P. Zhang, "A deep convolutional coupling network for change detection based on heterogeneous optical and radar images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 3, pp. 545–559, Dec. 2018.

[6] J.-F. Zhang, L.-L. Xie, and X.-X. Tao, "Change detection of earthquake-damaged buildings on remote sensing image and its application in seismic disaster assessment," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, vol. 4, Jul. 2003, pp. 2436–2438.

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[7] B. Liu, X. Li, and G. Zheng, "Coastal inundation mapping from bitemporal and dual-polarization SAR imagery based on deep convolutional neural networks," *J. Geophys. Res., Oceans*, vol. 124, no. 12, pp. 9101–9113, 2019.

[8] R. Manonmani and G. M. D. Suganya, "Remote sensing and GIS application in change detection study in urban zone using multi temporal satellite," *Int. J. Geomatics Geosci.*, vol. 1, no. 1, pp. 60–65, 2010.

[9] F. Gao, J. Dong, B. Li, and Q. Xu, "Automatic change detection in synthetic aperture radar images based on PCANet," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 12, pp. 1792–1796, Dec. 2016.

[10] M. Gong, X. Niu, P. Zhang, and Z. Li, "Generative adversarial networks for change detection in multispectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2310–2314, Nov. 2017.

[11] J. Liu et al., "Convolutional neural network-based transfer learning for optical aerial images change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 1, pp. 127–131, 2020.

[12] M. Yang, L. Jiao, F. Liu, B. Hou, and S. Yang, "Transferred deep learning-based change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6960–6973, Sep. 2019.

[13] P. Coppin, I. Jonckheere, K. Nackaerts, B. Muys, and E. Lambin, "Digital change detection methods in ecosystem monitoring: A review," *Int. J. Remote Sens.*, vol. 25, no. 9, pp. 1565–1596, 2004.

[14] J. Kittler and J. Illingworth, "Minimum error thresholding," *Pattern Recognit.*, vol. 19, no. 1, pp. 41–47, Jan. 1986.

[15] Z. Yetgin, "Unsupervised change detection of satellite images using local gradual descent," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 5, pp. 1919–1929, May 2012.

[16] S. Krinidis and V. Chatzis, "A robust fuzzy local information C-means clustering algorithm," *IEEE Trans. Image Process.*, vol. 19, no. 5, pp. 1328–1337, May 2010.

[17] M. Gong, Z. Zhou, and J. Ma, "Change detection in synthetic aperture radar images based on image fusion and fuzzy clustering," *IEEE Trans. Image Process.*, vol. 21, no. 4, pp. 2141–2151, Apr. 2012.

[18] N. A. Quarmby and J. L. Cushnie, "Monitoring urban land cover changes at the urban fringe from SPOT HRV imagery in south-east England," *Int. J. Remote Sens.*, vol. 10, no. 6, pp. 953–963, 1989.

[19] P. J. Howarth and G. M. Wickware, "Procedures for change detection using Landsat digital data," *Int. J. Remote Sens.*, vol. 2, no. 3, pp. 277–291, 1981.

[20] F. Bovolo and L. Bruzzone, "A theoretical framework for unsupervised change detection based on change vector analysis in the polar domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 1, pp. 218–236, Jan. 2007.

[21] J. S. Deng, K. Wang, Y. H. Deng, and G. Qi, "PCA-based land-use change detection and analysis using multitemporal and multisensor satellite data," *Int. J. Remote Sens.*, vol. 29, no. 16, pp. 4823–4838, 2008.

[22] A. A. Nielsen, K. Conradsen, and J. J. Simpson, "Multivariate alteration detection (MAD) and MAF postprocessing in multispectral, bitemporal image data: New approaches to change detection studies," *Remote Sens. Environ.*, vol. 64, no. 1, pp. 1–19, Apr. 1998.

[23] A. A. Nielsen, "The regularized iteratively reweighted mad method for change detection in multi- and hyperspectral data," *IEEE Trans. Image Process.*, vol. 16, no. 2, pp. 463–478, Feb. 2007.

[24] M. Liu, Q. Shi, A. Marinoni, D. He, X. Liu, and L. Zhang, "Super-resolution-based change detection network with stacked attention module for images with different resolutions," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Jul. 2022, Art. no. 4403718.

[25] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Jul. 2022, Art. no. 5607514.

[26] H. Chen, W. Li, and Z. Shi, "Adversarial instance augmentation for building change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Mar. 2022, Art. no. 5603216.

[27] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, Feb. 2022, Art. no. 8007805.

[28] H. Chen, C. Wu, B. Du, L. Zhang, and L. Wang, "Change detection in multisource VHR images via deep Siamese convolutional multiple-layers recurrent neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2848–2864, Apr. 2020.

[29] M. Zhang and W. Shi, "A feature difference convolutional neural network-based change detection method," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7232–7246, Oct. 2020.

[30] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, 2020.

[31] T. Liu et al., "Building change detection for VHR remote sensing images via local-global pyramid network and cross-task transfer learning strategy," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–7, 2021.

[32] D. Peng, L. Bruzzone, Y. Zhang, H. Guan, H. Ding, and X. Huang, "SemiCDNet: A semisupervised convolutional neural network for change detection in high resolution remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5891–5906, Jul. 2021.

[33] S. Tian, Y. Zhong, A. Ma, and L. Zhang, "Three-dimensional change detection in urban areas based on complementary evidence fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022, Art. no. 5608913.

[34] X. Jiang, G. Li, Y. Liu, X.-P. Zhang, and Y. He, "Change detection in heterogeneous optical and SAR remote sensing images via deep homogeneous feature fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1551–1566, 2020.

[35] H. Lee et al., "Local similarity Siamese network for urban land change detection on remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4139–4149, 2021.

[36] Q. Xu, K. Chen, X. Sun, Y. Zhang, H. Li, and G. Xu, "Pseudo-Siamese capsule network for aerial remote sensing images change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, Art. no. 6000405.

[37] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0924271620301532

[38] R. Liu, D. Jiang, L. Zhang, and Z. Zhang, "Deep depthwise separable convolutional network for change detection in optical aerial images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1109–1118, 2020.

[39] Y. Zhan, K. Fu, M. Yan, X. Sun, H. Wang, and X. Qiu, "Change detection based on deep Siamese convolutional network for optical aerial images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1845–1849, Oct. 2017.

[40] M. Zhang, G. Xu, K. Chen, M. Yan, and X. Sun, "Triplet-based semantic relation learning for aerial remote sensing image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 2, pp. 266–270, Feb. 2019.

[41] F. Schroff, D. Kalenichenko, and J. Philbin, "FaceNet: A unified embedding for face recognition and clustering," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 815–823.

[42] R. Caye Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[43] L. Mou, L. Bruzzone, and X. X. Zhu, "Learning spectral–spatial–temporal features via a recurrent convolutional neural network for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 924–935, Feb. 2019.

[44] R. Hadsell, S. Chopra, and Y. LeCun, "Dimensionality reduction by learning an invariant mapping," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, vol. 2, Jul. 2006, pp. 1735–1742.

[45] X. Qu, F. Gao, J. Dong, Q. Du, and H.-C. Li, "Change detection in synthetic aperture radar images using a dual-domain network," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, Art. no. 4013405.

[46] X. Tang et al., "An unsupervised remote sensing change detection method based on multiscale graph convolutional network and metric learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2021.

[47] Y. Sun, L. Lei, X. Li, X. Tan, and G. Kuang, "Patch similarity graph matrix-based unsupervised remote sensing change detection with homogeneous and heterogeneous sensors," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 4841–4861, Jun. 2021.

[48] Y. Sun, L. Lei, D. Guan, and G. Kuang, "Iterative robust graph for unsupervised change detection of heterogeneous remote sensing images," *IEEE Trans. Image Process.*, vol. 30, pp. 6277–6291, 2021.

[49] D. Xue, T. Lei, X. Jia, X. Wang, T. Chen, and A. K. Nandi, "Unsupervised change detection using multiscale and multiresolution Gaussian-mixture-model guided by saliency enhancement," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1796–1809, 2021.

[50] S. Saha, F. Bovolo, and L. Bruzzone, "Building change detection in VHR SAR images via unsupervised deep transcoding," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 1917–1929, Mar. 2021.

[51] S. Saha, Y. T. Solano-Correa, F. Bovolo, and L. Bruzzone, "Unsupervised deep transfer learning-based change detection for HR multispectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 856–860, May 2021.

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[52] Z.-G. Liu, Z.-W. Zhang, Q. Pan, and L.-B. Ning, "Unsupervised change detection from heterogeneous data based on image translation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Jul. 2022, Art. no. 4403413.
[53] C. Ren, X. Wang, J. Gao, X. Zhou, and H. Chen, "Unsupervised change detection in satellite images with generative adversarial network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10047–10061, Dec. 2021.
[54] C. Wu, H. Chen, B. Du, and L. Zhang, "Unsupervised change detection in multitemporal VHR images based on deep kernel PCA convolutional mapping network," *IEEE Trans. Cybern.*, vol. 52, pp. 1–15, 2021.
[55] T. Zhan, M. Gong, X. Jiang, and M. Zhang, "Unsupervised scale-driven change detection with deep spatial–spectral features for VHR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5653–5665, Aug. 2020.
[56] M. Gong, Y. Yang, T. Zhan, X. Niu, and S. Li, "A generative discriminatory classified network for change detection in multispectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 1, pp. 321–333, Jan. 2019.
[57] N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Trans. Syst., Man, Cybern. Syst.*, vol. SMC-9, no. 1, pp. 62–66, Feb. 1979.
[58] S. Li, H. Liu, Y. Gan, X. Zhu, and J. Zhang, "Image dehazing network based on residual dense block and attention mechanism," *J. Hunan Univ. Natural Sci.*, vol. 48, no. 6, pp. 112–118, 2021.
[59] S. Woo, J. Park, J. Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 3–19.
[60] F. Bovolo, L. Bruzzone, and M. Marconcini, "A novel approach to unsupervised change detection based on a semisupervised SVM and a similarity measure," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 7, pp. 2070–2082, Jul. 2008.
[61] C. Benedek and T. Szirányi, "Change detection in optical aerial images by a multilayer conditional mixed Markov model," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 10, pp. 3416–3430, Oct. 2009.
[62] I. Sutskever, J. Martens, G. E. Dahl, and G. E. Hinton, "On the importance of initialization and momentum in deep learning," in *Proc. Int. Conf. Mach. Learn.*, 2013, pp. 1139–1147.
[63] X. Glorot and Y. Bengio, "Understanding the difficulty of training deep feedforward neural networks," in *Proc. Int. Conf. Artif. Intell. Statist.*, 2010, pp. 249–256.



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